

Intelligent Student Performance Analytics and Learning Strategy Optimization: A Data Mining Approach

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Abstract

Predicting students at risk of academic failure and analyzing causes is critical when it comes to decision making stages. This study applies data mining techniques to the Open University Learning Analytics Dataset (OULAD) which contains around 32,000 data records with demographic, assessment and virtual learning environment (VLE) engagement data. This construct an analytical pipeline including data preparation, exploratory data analytics (EDA), supervised classifications and unsupervised learning. Under supervised classification this uses Random Forest and Gradient Boosting predictive models to predict at-risk students, achieving ROC-AUC scores of 0.9805 and 0.9822, respectively. Under unsupervised learning approach, K-Means clustering revealed three distinct student categories, while association rule mining (ARM) uncovers behavioral patterns in the dataset. The findings highlight that the performance related matrices such as total weighted score, assessment completion rate and submission counts are strong predictors in the analysis. The conclusion includes data driven recommendations for early warning system and academic support.

Keywords: Data Mining, Student Performance Analysis, Clustering, Association Rule Mining

1 Introduction

In the worldwide higher education sector, student retention and academic success are challenging to maintain. Early identification of students who have the chance to fail or withdraw enables the institute to get necessary actions timely and for targeted students. In current traditional approaches, educators are mainly relying on reactive

indicators such as mid-term marks, etc. This may not be effective, since there is no reasonable time for effective remediation.

Data Mining provides a proactive alternative approach which creates predictive models from historical data. With the effect of digitization, there are data sources that can be used to collect data, which is helpful for this approach. This study aims to perform comprehensive analysis to figure out the factors influencing student outcomes. Then it builds and evaluates supervised predictive models to identify at-risk students. After that it applies grouping methods and association rules to find natural grouping and association among students. Finally, from the outcomes this suggests actionable recommendations for educators and institutions.

Throughout the analysis it uses Open University Learning Analytics Dataset which contains a multi-source view of student demographics, assessment submissions, VLE interactions and registration behaviors across multiple course presentations.

2 Data Description and Preparation

2.1 Dataset Overview

The OULAD contains seven CSV files which are interrelated. Each file contains student information, assessment records, VLE activity logs, course metadata, and registration data. After evaluating other available options, I decided to go with OULAD because of its scale, richness and enabling feature construction from multiple sources.

2.2 Data Integration Pipeline

To construct a unified master data frame this went through a multi stage merging pipeline.

- Assessment Aggregation: Assessment metadata was merged with individual student submissions. Then created an insightful column called "weighted score", which computed as $(\text{score} \times \text{weight}) / 100$. After that it aggregates per student to derive "total weighted score", "average score" and "assessments submitted".

- VLE Engagement Aggregation: VLE resource metadata was merged with student interaction logs. Then it aggregates per student to derive "total clicks", "unique activities", "active days" and "total vle interactions".

- Registration Features: Registration data aggregated per student to derive "earliest registration", "has unregistered", "number of modules registered" and "number of modules unregistered".

- Master dataset construction: The student info table is considered as the base since it contains more valuable info about each student. The three aggregated datasets were left joined on student ID. This creates the final dataset of 32,593 rows and 23 columns.

2.3 Missing Value Treatment

While analyzing the null values in the final dataset, there were 5847 null values in the assessment aggregation related columns, 2852 null values in the VLE engagement aggregation related columns and 32 registration date related columns. Those values were filled with 0 since it reflects genuine reasoning for each of them.

3 Exploratory Data Analysis

3.1 Target Variable Distribution

Initially the "final result" values had clear class imbalance: Pass (12,361), Withdrawn (10,156), Fail (7,052), and Distinction (3,024). In the newly created "is at risk" binary variable, it was balanced by considering a group of Fail and Withdrawn students are at risk (17208) and other groups of students (Pass and Distinction) are not at risk (15385).

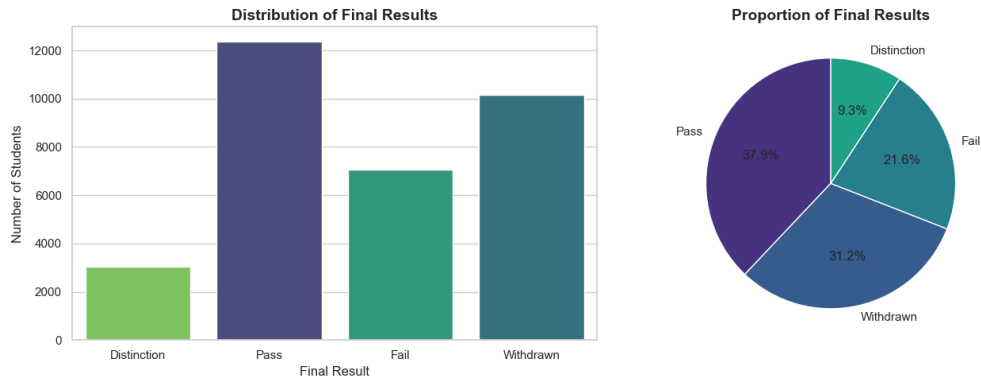


Fig. 1: Distribution of final results across the student population.

3.2 Demographic Analysis

Here it analyzed the final result with age band and gender. It reveals that the younger age band (0-35) has both a high number of Pass and Withdrawn student counts. This seems natural since most students who are studying belong to this age band. Gender differences are generally modest. Males show slightly higher values of withdrawals.

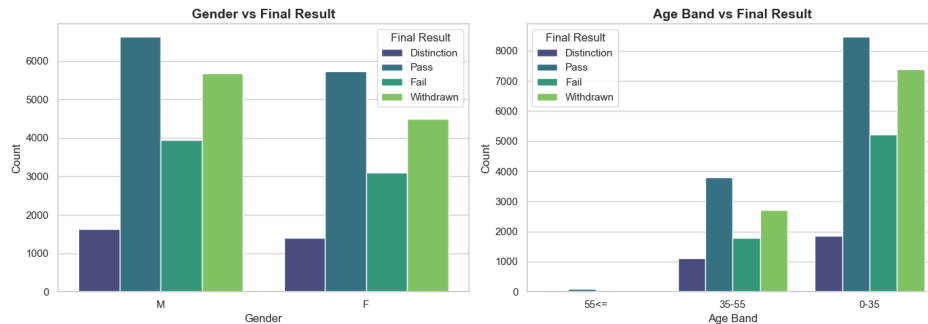


Fig. 2: Final result distribution by gender and age band.

Educational analysis reveals that the student has more prior qualifications and exhibits a higher pass rate than others. Disability status also reveals that students who are disabled have a higher withdrawal rate.

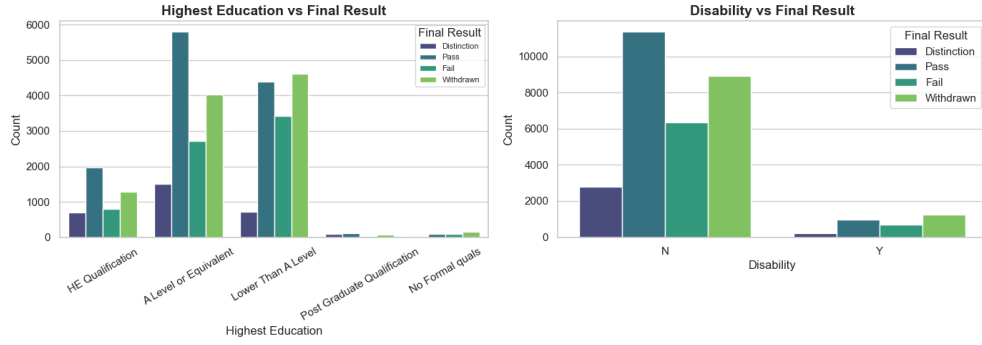


Fig. 3: Final result distribution by highest education level and disability status.

3.3 Engagement and Assessment Patterns

Box plots in VLE engagement reveal that successful students exhibit substantially higher total clicks, active days, unique activities, and total VLE interactions compared to at-risk students.

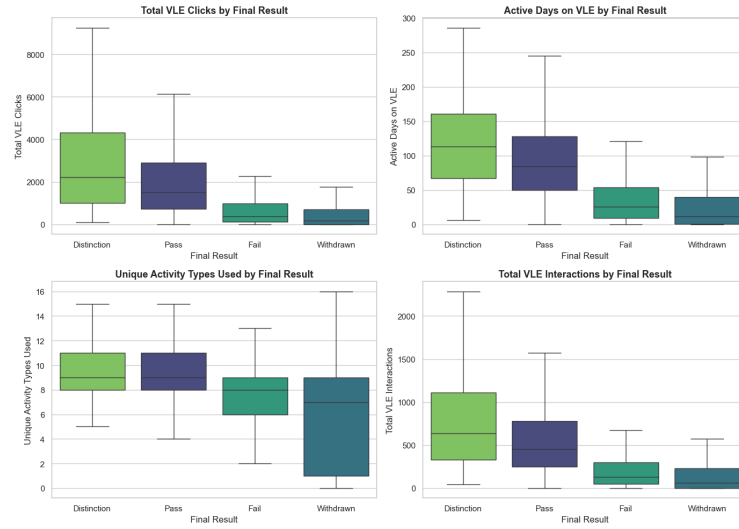


Fig. 4: Distribution of VLE engagement metrics across final result categories.

Assessment performance metrics follow a similar pattern. Higher average score, higher total weighted score and higher number of assessment submissions directly correlate with success students.

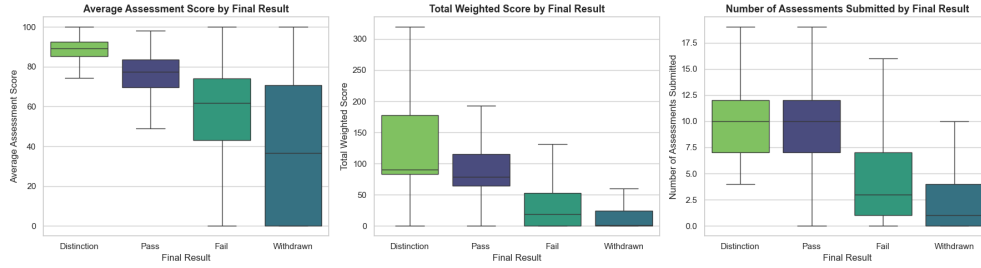


Fig. 5: Assessment performance metrics across final result categories.

3.4 Feature Correlations

The correlation heatmap is created to get an understanding about feature correlation. It reveals higher correlation between VLE engagement features (total clicks, active days, unique activities, total vle interactions) and between assessment features (avg score, total weighted score, assessments submitted). The correlation in between those two domains are moderate. This suggests both of those domains contribute complementary information.

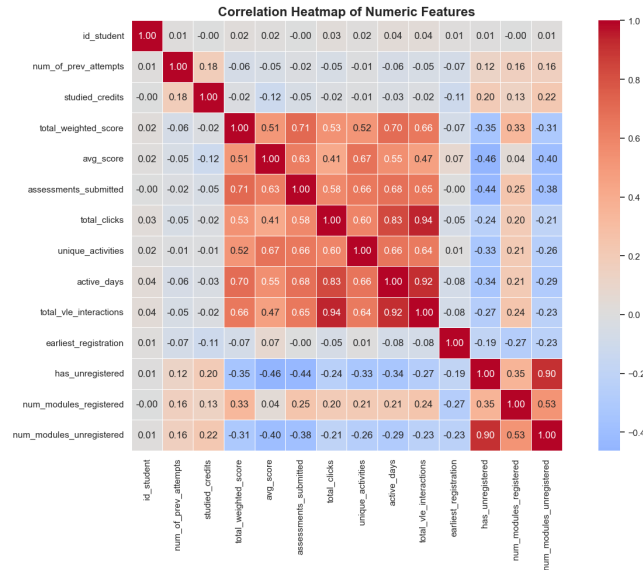


Fig. 6: Correlation heatmap of numeric features.

There are few more EDA steps available in the notebook.

4 Feature Engineering

To capture higher level behavioral patterns few engineered features were created.

- "is at risk": This is the binary target variable created by assigning 1 for Fail/Withdrawn results and 0 for Pass/Distinction results.
- "clicks per day": This was created to get engagement intensity. This is computed as total clicks divided by active days.
- "assessment completion rate": This calculates the ratio of assessments submitted out of total available assessment count.
- "score per assessment": This is calculated by dividing the total weighted score from the number of assessments submitted.
- "early registration": This checks the students who have enrolled the course more than 100 days before and adds a binary flag.

After creating these new features, I looked into the categorical columns within the final dataset. Mainly it contained 8 categorical columns: gender, region, highest education, IMD band, age band, disability, code module and code presentation. These were label encoded. At the end the dataset contains 24 features which covers areas such as Demographic features, Academic context, Assessment performance features, VLE engagement features, Registration features and Engineered features.

5 Predictive Modeling

5.1 Experimental Setup

The data set is divided into an 80/20 ratio which contains 26074 samples in the training set and 6519 samples in the test set. In both of these sets, the proportion of students at risk is the same. Therefore, I decided to go with this split. The features were standardized using StandardScaler and two classifiers were trained and evaluated.

5.2 Random Forest Classifier

A Random Forest with 200 estimators, 15 max depth and balanced class weights were defined to train the model. The model results are as follows.

Metric	Value
Accuracy	0.9296
F1-Score	0.9319
ROC-AUC	0.9805
5-Fold CV Mean F1	0.9309

Table 1: Random Forest Results

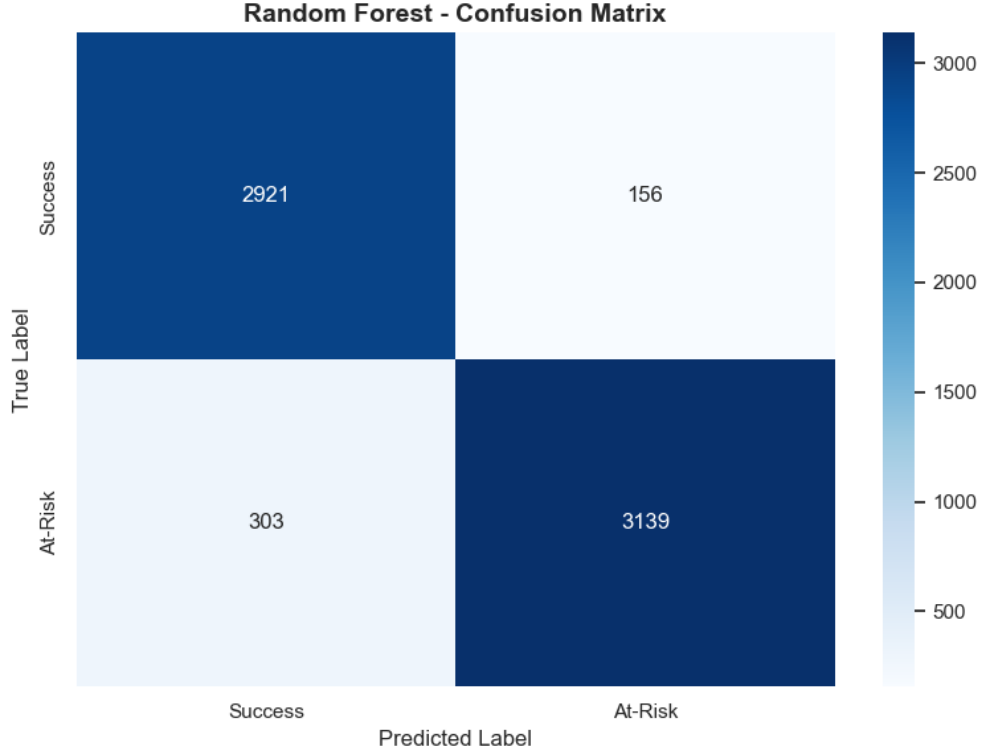


Fig. 7: Confusion matrix for the Random Forest classifier.

5.3 Gradient Boosting Classifier

A gradient boosting model with 200 estimators, 5 max depth and 0.1 learning rate were defined to train the model. The model results are as follows.

Metric	Value
Accuracy	0.9336
F1-Score	0.9357
ROC-AUC	0.9822
5-Fold CV Mean F1	0.9340

Table 2: Gradient Boosting Result

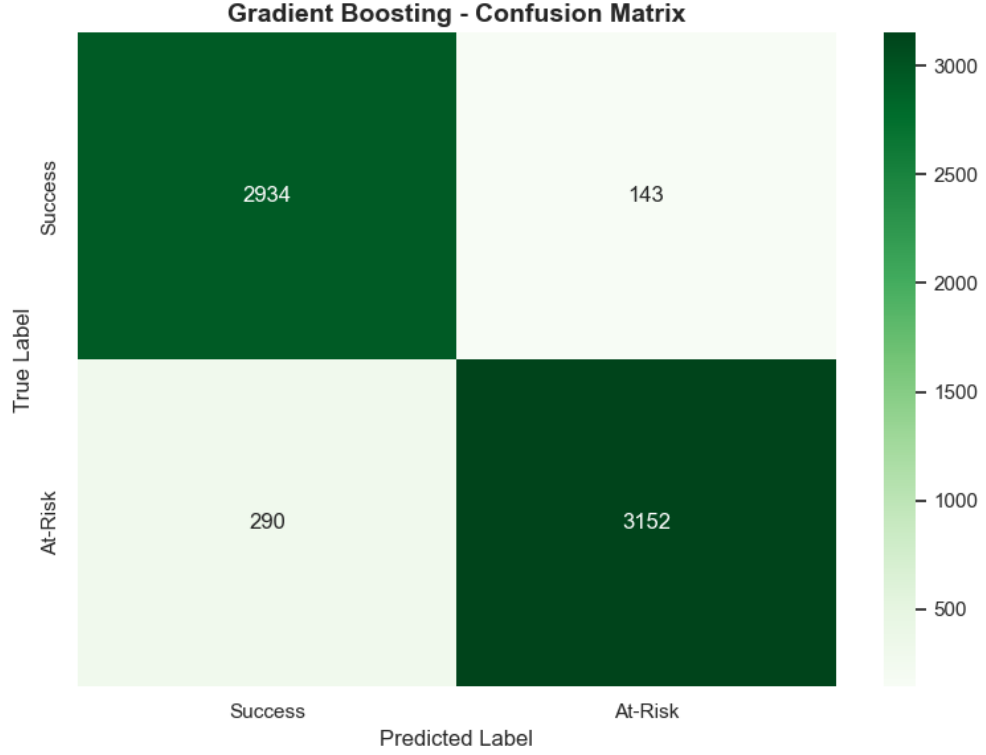


Fig. 8: Confusion matrix for the Gradient Boosting classifier.

5.4 Model Comparison

According to all metrics Gradient Boosting is marginally outperforming Random Forest. ROC-AUC plot is used to demonstrate how well these two models perform classifications. Both models demonstrate excellent discriminative ability with ROC-AUC scores above 0.98.

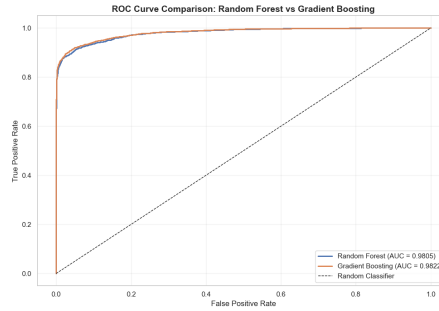


Fig. 9: ROC curve comparison of Random Forest and Gradient Boosting classifiers.

5.5 Feature Importance Analysis

When comparing models, both agree that performance related features like "total weighted score", "assessment completion rate" and "assessment submitted" are dominating predictive power. Registration behaviors like "has unregistered" too have some significant power. VLE engagement features such as active days, total clicks provide average meaningful predictive value.

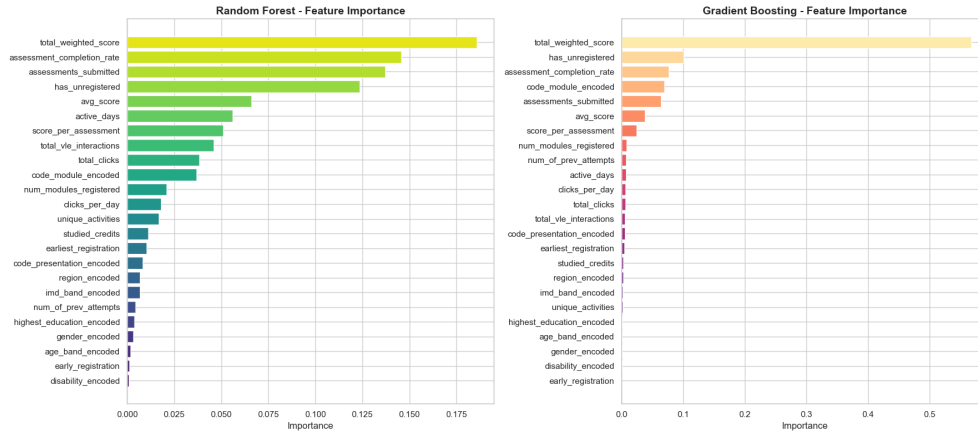


Fig. 10: Feature importance rankings for RF and GB models.

6 Unsupervised Learning

6.1 K-Means Clustering

6.1.1 Optimal Cluster Selection

To figure out the optimal number of clusters this uses the Elbow Method along with Silhouette Score analysis across K=2 to K=10. It clearly gives K=3 is the optimal number with the highest silhouette score of 0.351.

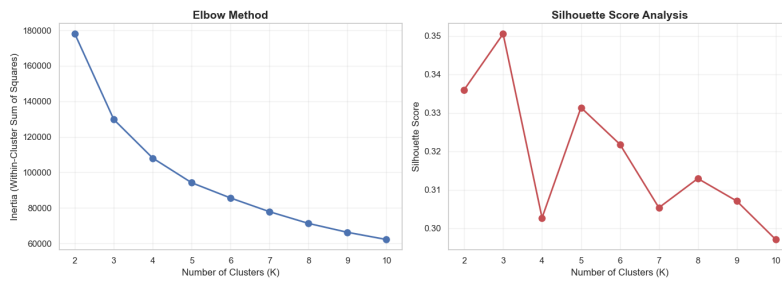


Fig. 11: Elbow method and silhouette score analysis for cluster selection.

6.1.2 Cluster Profiles

Feature	Cluster 0 (Moderate)	Cluster 1 (Disengaged)	Cluster 2 (High Performers)
Total Clicks	905.97	69.66	4239.72
Active Days	52.46	4.74	147.07
Avg Score	72.17	6.94	79.46
Total Weighted Score	54.18	0.69	133.89
Assessments Submitted	6.49	0.29	11.86

Table 3: Cluster profiles with mean values

Cluster vs Final results confirm:

- Cluster 1 (Disengaged): 74% Withdrawn, 26% Fail. Virtually all at-risk.
- Cluster 2 (High Performers): 62% Pass, 22% Distinction. Predominantly successful.
- Cluster 0 (Moderate): 43% Pass but 26% Fail. Mixed outcome.

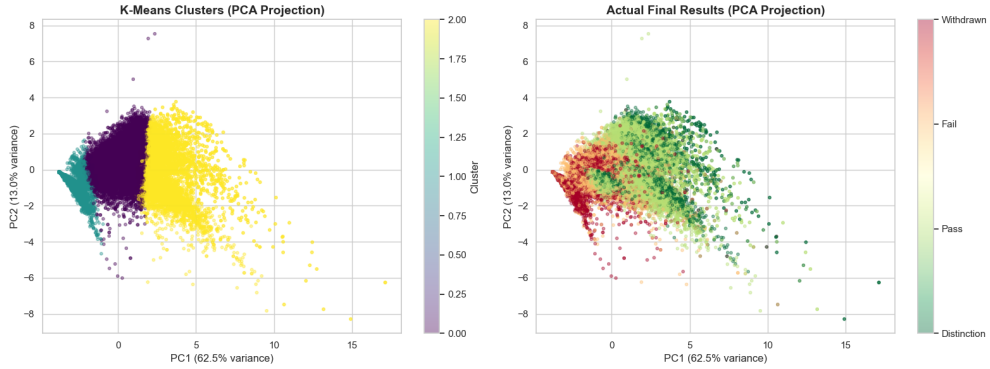


Fig. 12: PCA projection of K-Means clusters (left) and actual final results (right).

6.2 Association Rule Mining

Apriori algorithm with minimum support of 0.05 is set to find frequent itemset and association rules. For the numerical columns, first create meaningful bins and use one hot encoding. With this encoding there are 32 binary features used for rule mining. Along with this setup it yielded 1,655 frequent itemsets and 7,320 association rules (minimum confidence 0.5). Of these, 1,400 rules specifically predict final result categories. Key high lift rules include low score, VLE inactive, few assessments, disability can lead to withdrawal.

This helps to keep a next step beyond the supervised learning findings which provides interpretable behavioral patterns that educators can directly act upon.

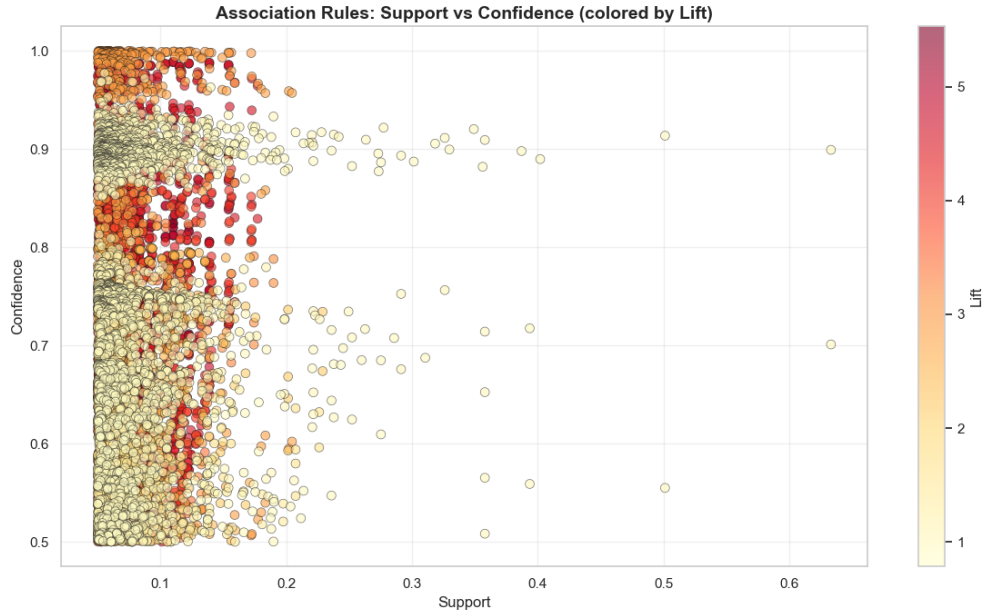


Fig. 13: Association rules: support vs. confidence, colored by lift.

7 Discussion

7.1 Key Insights

These are a set of key insights that come along with the analysis.

- Assessment Engagement is the strongest predictor of students at risk. In the feature importance rank, total weighted score alone accounts for 18.6% and 56.7% in the Random Forest and Gradient Boosting respectively.

- Registration behavior provides an early warning signal. The "has unregistered" feature is ranked in 4th and 2nd positions in Random Forest and Gradient Boosting importance rankings respectively. This suggests that students who unregistered modules exhibit early disengagement.

- Clustering reveals specific three student groups. A disengaged group with near zero participation and almost certain failures, a high performing group with consistent engagement with high scores, and a critical moderate group that shows engagement but mixed outcomes. This is the specific sector that needs high attention.

- VLE Engagement is good to have, but it doesn't replace the effect of assessment based features. This suggests that passive browsing is less indicative than active assessment participation.

7.2 Ethical Considerations

Predictive models used to identify at-risk students carry ethical implications.

- Fairness: Models must be audited based on demographic biases. To train models this uses protected characteristics such as gender, disability status, age band etc. The student should flag based on their study habits, grades, and attendance. If the model is flagging students more often simply because they belong to a certain demographic group, even if their grades are actually good, that is a kind of bias prediction we should audit.

- Privacy and Consent: While using these models in action it needs to collect personal data from the student. These data should be handled in a regulated manner and should get the consent from the student as well.

- Transparency: These data should be used to support students but not to penalize. There should always be human interaction in the decision making stage without automating the judging process since this can affect for student's feelings and future.

7.3 Recommendations

These are the set of recommendations based on analysis.

- Deploy an early warning system: Use either one of the above predictive models to flag students at-risk within the few weeks of a course, using assessment submission patterns and VLE engagement as primary signals.

- Target moderate engagement students: Investing time on these students (cluster 0) will give the best return of investment. Proactive approach is recommended for educators since if they were approached in the early stage they can move forward with a good result and reduce the risk of failing.

- Monitor registration behavior: Module unregistration should be taken as an early red flag. Administrators should keep an eye on those students and follow up with them. That will help to reduce the chance of getting low results for those students.

- Assessment completion support: Since assessment completion rate is among the top predictors of success, friendly reminders and helpful tips will keep students moving. If it helps them to finish their assignments, the above analysis says, they are much more likely to pass the class.

- Investigate disengaged students: Students belong to cluster 1 monitored here. This helps to understand barriers to participation, such as personal circumstances, lack of preparedness, or course fit issues.

8 Conclusion

This study mainly focused on demonstrating the effectiveness of data mining techniques for student performance. This is based on the OULAD dataset and it uncovers actionable patterns in it. A comprehensive analytical pipeline was created from data cleaning, feature engineering, predictive modeling, clustering, association rule mining to data driven decision making. Both supervised models achieved great performance reporting ROC-AUC greater than 0.98. Assessment engagement related features dominated the predictive importance. Clustering method revealed three distinct student profiles with clear differences. These findings provide foundation for implementing data driven early warning systems and personalized learning support in higher education.

Appendix A Notebook

The complete notebook for this analysis is available in <https://github.com/SDPcodes/student-performance-analytics>

References

[1] OULAD (Open University Learning Analytics Dataset) <https://archive.ics.uci.edu/dataset/349/open+university+learning+analytics+dataset>